

Shobitha A Rao

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Genetics postgraduate with a keen interest in Molecular Biology, Genomics, and Bio-informatics. Skilled in laboratory techniques, Genetic analysis, Data interpretation, and Scientific research. Looking for opportunities to apply my knowledge and contribute to research that involves molecular mechanisms, genomics, and biotechnological innovation.

SKILLS

Molecular Biology

DNA & RNA Isolation, PCR, Agarose Gel Electrophoresis, Gel Prep & Doc, Primer Handling, NanoDrop

Protein Analysis

SDS-PAGE, Western Blotting, Protein Extraction & Quantification

Bioinformatics

FastQC, fastp, MultiQC, BLAST, QUAST, BUSCO, MaSuRCA, SPAdes, SOAPdenovo2, Augustus, Linux/Unix

Lab Techniques

Centrifugation, Micropipetting, Buffer/Reagent Prep, GLP

Software Skills

MS Office, Google Workspace, Notebook LM

Languages

English, Hindi, Kannada (Fluent)

Soft Skills

Teamwork, Presentation, Communication, Time Management, Leadership, Problem Solving

EDUCATION

Masters in Genetics

SRM Institute of Science and Technology, Kattankulatham
2024 – 2026
CGPA: 7.5

Bachelors in Biotechnology

NMAMIT, University in Nitte, Karnataka
2020 – 2024
CGPA: 6.8

ACADEMIC PROJECTS

Whole Genome Sequencing and De Novo Assembly of *Caesalpinia bonducella* L.

- Performed de novo whole genome sequencing and assembly of *Caesalpinia bonducella* using MaSuRCA.
- Successfully assembled and characterized the chloroplast and mitochondrial organelle genomes using GetOrganelle.
- Conducted gene prediction using AUGUSTUS, identifying 15,052 genes, and performed functional annotation via BLAST against the NCBI nr database, revealing 8,765 conserved genes and 6,287 potentially novel genes unique to the species.
- Ensured high-quality sequencing input by isolating genomic DNA with OD A260/280 values of 1.87–1.89, and validated raw reads using FastQC.

Study on Efficacy of Hydrogels in Soil Water Retention and Plant Growth

- Synthesized biodegradable hydrogel beads using sodium alginate with natural gums (guar gum, locust bean gum, bacterial cellulose), achieving a water-holding capacity of 82% and re-swelling capacity of 83.33%.
- Demonstrated that SLGC hydrogels improved germination and sustained growth in radish, green gram, paddy, and spinach under reduced watering conditions.
- Conducted systematic soil water retention studies including swelling ratio (72%), water retention (78.95%), and pH stability analysis for hydrogel performance evaluation.
- Validated hydrogels as sustainable soil conditioners through tray and pot studies under water-deficit conditions.

LABORATORY EXPERIENCE

- Assisted in sample preparations, data collection, and quality control experiments.
- Operated and calibrated lab instruments to ensure precision.
- Conducted literature reviews supporting ongoing research publications.
- Followed SOPs, documented results, and maintained laboratory logs accurately.

CERTIFICATIONS & ACTIVITIES

- **International Conference on Bioengineering, SRM** – Student Support Team Member
- **Genetic Engineering Research Symposium, SRM** – Poster Presentation
- **Internship at NUCSER** – July 2023

PUBLICATIONS

Biopolymeric Hydrogel as an Eco-Friendly Soil Amendment for Water Conservation and Sustainable Agricultural Applications.

May 2026

International Journal of Aquatic Research and Environmental Studies, 6(S1), 493-516

Authors: D. S, C. S Tantry, D. M. Chandy, E. R. Nyirenda, S. Shetty, Z. Shaikh, **Shobitha. A. Rao**, M. B. Krishnaswamy, M. Kumar G T, V. S. Muddappa

[View Publication](#)